PE-RankFormer: A Minimally-Mechanistic Relational Transformer Matches a Strongly Mechanistic Model for Prime-Editing Efficiency Prediction

PEFormer research pilot

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Abstract

Prime editing (PE) efficiency prediction has been dominated by mechanistic models that hard-code assumptions about the PE reaction pathway (nick, flap resolution, mismatch repair, etc.), most recently OptiPrime (Hsu et al., 2026), a strongly mechanistic model trained on 297,962 PE experiments. We ask whether a minimally-mechanistic, general-purpose relational Transformer, given the same training data, can match or exceed it. We implement PE-RankFormer – a 26.6M-parameter paired-sequence Transformer with cross-attention between the edit-outcome encoding and the pegRNA encoding and FiLM-based conditioning on experimental context – and add a 3-way categorical "simplex" outcome head that models the mutually exclusive unedited/edited/indel read fractions rather than regressing edited-read fraction alone.

The decisive experiment uses OptiPrime's actual training mix, obtained from its authors: the exact 297,962 training rows, their five cross-validation splits, and a held-out test set of 20,509 rows that neither model ever trains on. Both models are five-model cross-validation ensembles trained on the identical rows, matching OptiPrime's own protocol exactly. On the **complete held-out set** ($n = 20{,}509$), PE-RankFormer reaches Spearman $\rho = 0.8865$ against OptiPrime's 0.8690 – a difference of +0.0175 with a protospacer-clustered paired bootstrap 95% CI of $[+0.0068, +0.0279]$ and $p = 0.001$. **PE-RankFormer is ahead, with the difference outside chance.**

That result is not uniform across the two studies contributing to the held-out set. On the 9,175-row Liu partition – the paper's own primary held-out benchmark – the two models are statistically indistinguishable ($\rho = 0.8349$ vs. 0.8365, $p = 0.94$). The entire measured advantage is concentrated in the 11,334-row Kim partition ($\rho = 0.775$ vs. 0.732), which spans seven cell lines and multiple Cas9/PE-system variants absent from Liu. We report both results rather than the pooled number alone, since they support different claims: parity on the benchmark the paper itself emphasizes, and a measurable edge on a more experimentally diverse one.

We also report, and retract, an earlier result. Before the authors' data was available we reconstructed the corpus from public sources and measured an apparently decisive margin over OptiPrime (+0.070 to +0.083 Spearman, $p < 0.0005$). That margin was an artifact: lacking upstream genomic context for part of the corpus, we had left-padded sequences with filler bases before running OptiPrime, degrading its inputs. On correctly formatted data OptiPrime's Liu-only score rises to 0.8365 from 0.724, and the margin disappears. We document the reconstruction effort, the artifact, and the corrected conclusion in full, since the failure mode – a baseline handicapped by input preprocessing rather than by modelling – is easy to reproduce and hard to detect from aggregate metrics alone.

Contents

1 Introduction

Prime editing efficiency prediction is typically approached with strongly mechanistic models: pipelines that explicitly encode the PE reaction as a sequence of biochemical steps (spacer/PBS/RTT annealing, flap competition, mismatch-repair outcome) and use that structure to constrain the model. OptiPrime (Hsu et al., 2026) is the current state of the art on this axis, trained on 297,962 PE experiments assembled across four studies.

The research question motivating this pilot is whether a *minimally mechanistic* model – one that encodes only the raw sequences and experimental metadata and lets a general-purpose relational architecture (a Transformer with cross-attention between the edit and pegRNA representations) discover whatever structure is predictive – can match or outperform a strongly mechanistic model of comparable training scale. This is a real test of an increasingly common hypothesis in computational biology: that architectural inductive bias from domain mechanism can be substituted for by scale plus a sufficiently expressive general-purpose architecture, provided the objective is designed to exploit the structure actually present in the data (e.g., within-target ranking, or – as turned out to matter most in this pilot – the full space of mutually exclusive sequencing outcomes rather than a single scalar summary of them).

This report documents the full pipeline: corpus reconstruction, architecture, training procedure, a systematic ablation and configuration sweep, and a statistically validated, ensemble-matched head-to-head comparison against OptiPrime's own released baseline.

2 Training Data

The results in this report rest on OptiPrime's *actual* training mix, provided by its authors: 58 CSVs comprising the exact 297,962 training rows, their five cross-validation splits, a held-out test set of 20,509 rows, and a per-row `weight` column derived from read depth. Reading those files reproduces the published figure exactly.

Table 1: OptiPrime's official training mix, as supplied by the authors.

Partition	Training rows (splits 1–5)	Held-out test
Liu (Hsu 2026)	65,594	9,175
Kim (DeepPrime)	58,301	11,334
Schwank (PRIDICT / PRIDICT2.0)	174,067	0
Total	**297,962**	**20,509**

One line of that table resolves a question that consumed a large part of this project. The public Hsu supplementary workbook contains 74,769 Liu rows, but the paper's own partition table accounts for only 65,594, and we could not identify any quality filter that produced the difference. The answer is that 74,769 = 65,594 + 9,175: the "excess" was OptiPrime's held-out test set, and the 297,962 figure counts training rows only. **No filter ever existed.** Section 2.1 records the search, because every negative result in it was individually correct and the collective error was one of framing rather than of method.

Metadata absent from a file is inferred from its name, mirroring OptiPrime's own `scripts/pe/pe_datasets.py`. Their files lowercase the appended U6 transcription-start base (`gCAGG...`) where we uppercase it; that is the only discrepancy between their Liu designs and our independent extraction, which agree on 19,908 of 19,908 designs once case is normalised.

Indel coverage. `indel_frac` is present for the Liu rows and the smaller Schwank libraries but absent for all Kim rows and the Schwank diverse libraries – 42.5% of the corpus has no measured indel. Rather than impute zero, which would assert an unobserved measurement, the simplex head marginalises the indel class out for those rows and scores the remaining binary unedited-vs-edited decision (§3.1).

2.1 Independent reconstruction from public data

Before the authors' files were available we reconstructed the corpus from public releases, verified against a 42-partition ground-truth breakdown (by study × cell type × PE system) reported in the OptiPrime paper. That reconstruction reached 285,260 rows (95.7%) and is what the training runs in §5 used. It is documented here because it is the basis of the superseded results in §7, and because parts of it – notably the PRIDICT2.0 K562 derivation below – were independently validated and remain correct.

Table 2: Final reconstructed corpus composition (`data/processed/optiprime_full_297962.parquet`).

Source	Partitions matched (of target)	Rows
Liu (Hsu 2026, newly generated)	4/4 (raw count; see note)	74,769
Kim (DeepPrime)	18/18 exact	58,301
Schwank (PRIDICT / PRIDICT2.0)	19/20 exact	152,190
Total	**37/42 target-exact**	**285,260**

285,260 of 297,962 target rows (95.7%) were recovered with a documented, reproducible provenance trail. The largest single component, the 22,752-row PRIDICT2.0 K562 partition, was rebuilt directly from the PRIDICT2.0 release (`dataset/proc_v2/data_23k_v1.csv` and `k562_indices_nan.pkl`). That file ships no literal spacer/PBS/RTT columns, so all three are sliced out of the wide target sequences using the release's coordinate columns:

$$\begin{aligned} \text{spacer} &= \texttt{'G'} + \text{wide\_initial}[\text{protospacerlocation}] \\ \text{pbs} &= \text{revcomp}(\text{wide\_initial}[\text{PBSlocation}]) \\ \text{rtt} &= \text{revcomp}(\text{wide\_mutated}[\text{RT\_mutated\_location}]) \end{aligned}$$

where the leading `G` is the U6 transcription-start base. This derivation was validated against an independent reconstruction that read the same columns from the PRIDICT2.0 supplementary spreadsheet: all 22,734 of its rows are reproduced, with `edited` values agreeing to 5×10^{-11}.

Two remaining discrepancies were investigated and ruled out rather than papered over: (1) the Hsu/Liu partition contains 74,769 raw rows against a 65,594-row target with no public QC/coverage column to explain the gap. Several plausible filters (PE2/PE4 co-presence, barcode-collision exclusion, invalid-read exclusion, zero-efficiency exclusion, OptiPrime's own embedding-size bounds) were each tested and rejected; OptiPrime's released data loader applies no row filter and its input schema carries no QC column; the article publishes no Source Data; and the reported efficiencies are replicate averages, not raw read ratios, so no read-depth threshold can be back-computed. The filter provably lived in an internal pipeline upstream of everything released, and we use the full 74,769. (2) The Schwank K562 PE4+epegRNA partition (target 21,201) is absent from any public release, including all four cited BioProjects at the SRA metadata level. A residual ∼1,611 rows (subscreen/Library2 fractions of two partitions) are left unclaimed rather than risk mis-assignment, as the subscreen files carry no sequence columns. Full detail in `reports/training_data_reconstruction.md`.

Cross-validation folds were built to be **protospacer-disjoint**: fold 0 (test), fold 1 (val), folds 2–4 (train), seed 20260812, with zero-leakage verified (no protospacer appears in more than one fold).

3 Model Architecture

PE-RankFormer is a 26.6M-parameter Transformer (d_{model}=384, 6 attention heads, FFN width 1536, dropout 0.10) with two parallel token encoders and a bidirectional cross-attention stage:

- **Edit encoder** (6 layers): a paired-sequence encoding of the unedited and edited full target sequence, so the model can directly attend to what changed rather than being told a structured description of the edit.
- **PegRNA encoder** (4 layers): a segment-aware encoding of the spacer, PBS, and RTT, with segment-type embeddings distinguishing the three regions.
- **Cross-attention** (2 blocks): bidirectional cross-attention between the edit-token and pegRNA-token representations, followed by attention pooling on each side.
- **FiLM conditioning**: pooled representations are modulated by an embedding of experimental context (cell type, PE system variant, Cas9/PAM variant, scaffold, motif), so the same sequence pair can receive different predictions in different experimental contexts.
- **Prediction head**: either a scalar sigmoid head (edited-fraction regression) or the 3-way simplex head described in §3.1.

No PE-reaction-specific mechanism (e.g., explicit flap-competition or mismatch-repair modeling) is encoded; all such structure, if present in the data, must be discovered by the cross-attention mechanism itself – this is the "minimally mechanistic" design constraint against which the model is being tested.

3.1 The simplex outcome head: diagnosis-driven innovation

Early training runs (Model B, scalar head, no ranking loss) showed clear overfitting: training loss fell roughly $4\times$ over 30 epochs while validation Spearman ρ plateaued from epoch $\sim$20 onward (Figure 1). Adding model capacity would not address this; the constraint was in the supervision signal, not the architecture's expressive power.

Every PE sequencing read falls into exactly one of three mutually exclusive categories: *unedited*, *edited* (the intended outcome), or *indel* (an unintended byproduct of the nick/repair process). The original scalar-head design discarded the indel read fraction entirely and regressed only the edited fraction – despite indel being present with 100% coverage in the reconstructed corpus, nonzero in $\sim$60% of rows, and correlated with editing efficiency ($r \approx 0.25$). This is a real, cheaply available signal being thrown away.

We replaced the scalar sigmoid head with a **3-way simplex head**: a softmax over (unedited, edited, indel) logits trained with soft cross-entropy against the observed read proportions (clipped/renormalized on the rare rows where edited+indel exceeds 1 due to counting noise). At inference, efficiency = $\mathrm{softmax}(\mathrm{logits})_{\mathrm{edited}}$, and the ranking score used for the pairwise ranking loss is the edited-vs-unedited log-odds, $\mathrm{logit}_{\mathrm{edited}} - \mathrm{logit}_{\mathrm{unedited}}$.

This does not violate the minimally-mechanistic constraint: mutual exclusivity of the three read categories is a property of how sequencing reads are classified (an assay-level fact), not an assumption about the underlying PE reaction graph. No biochemical step ordering or intermediate state is encoded.

The simplex head produced the top 4 single-model validation Spearman results of the entire sweep (§5). Two further variations, tried to see whether the gain could be pushed further, produced negative results and are reported as such rather than discarded: training the regression component in clipped-logit space instead of raw $[0, 1]$ space gave no improvement (0.866 vs. 0.864 raw, within seed noise), and increasing dropout to 0.20 gave no improvement (0.871 vs. 0.870 at dropout 0.10).

4 Training Procedure

Models are trained with joint regression (Huber loss) and pairwise RankNet within-target ranking loss (λ_{rank} weighting), using a custom grouped batch sampler that biases minibatches toward rows sharing a within-target ranking group – necessary because the $\sim$220,000 distinct ranking groups in the full corpus mean uniform random batching yields close to zero rankable pairs per batch. Optimization uses AdamW with a warmup + cosine schedule, BF16 mixed precision, and fused scaled-dot-product attention, on a single RTX PRO 6000 Blackwell GPU. All models in the final sweep were trained for a full 30 epochs (early-stopping patience matched across configurations after an initial unfair comparison was caught and corrected – see §8).

5 Ablation and Configuration Sweep

A bounded sweep over outcome head (scalar vs. simplex), regression target space (raw $[0, 1]$ vs. clipped-logit), ranking-loss weight λ_{rank}, context conditioning (on/off), and dropout was run, each configuration trained to full 30 epochs at matched patience. All selection decisions in this sweep were made on the **validation** fold only.

Table 3: Full configuration sweep, sorted by best validation Spearman ρ (single seed 20260812 unless noted). "rank" = with ranking loss (λ_{rank}=0.25 for Models A/C, 0.05 for the low-rank probe); blank = ranking loss off (λ_{rank}=0).

Run	Head	Space	Context	λ_{rank}	Val ρ
final_simplex_s2	simplex	raw	on	0.00	**0.879**
final_simplex_s3	simplex	raw	on	0.00	0.873
exp_dropout20_simplex	simplex	raw	on	0.00	0.871 (dropout 0.20)
exp_simplex	simplex	raw	on	0.00	0.870
model_b_norank	scalar	raw	on	0.00	0.864
final_logit_s3	scalar	logit	on	0.00	0.867
exp_logit_norank	scalar	logit	on	0.00	0.866
final_logit_s2	scalar	logit	on	0.00	0.862
exp_lowrank005	scalar	raw	on	0.05	0.850
model_a_rank	scalar	raw	on	0.25	0.786
model_c_nocontext_fair	scalar	raw	off	0.25	0.705
model_c_nocontext	scalar	raw	off	0.25	0.675 (early-stopped, unfair)

Three findings stand out. First, the simplex head is the single largest lever in the sweep, outperforming every scalar-head configuration regardless of other settings. Second, the ranking loss *hurts* single-model validation Spearman at the weights tried (λ_{rank}=0.25 costs $\sim$0.08 ρ relative to λ_{rank}=0); this is a genuine trade-off against within-target ranking quality that a scalar global Spearman number does not fully capture (see §8), and is noted as a caveat rather than resolved in this pilot. Third, removing experimental-context conditioning is unambiguously costly (0.705 vs. 0.786 at matched patience, both with ranking loss on), confirming the FiLM conditioning mechanism is doing real work.

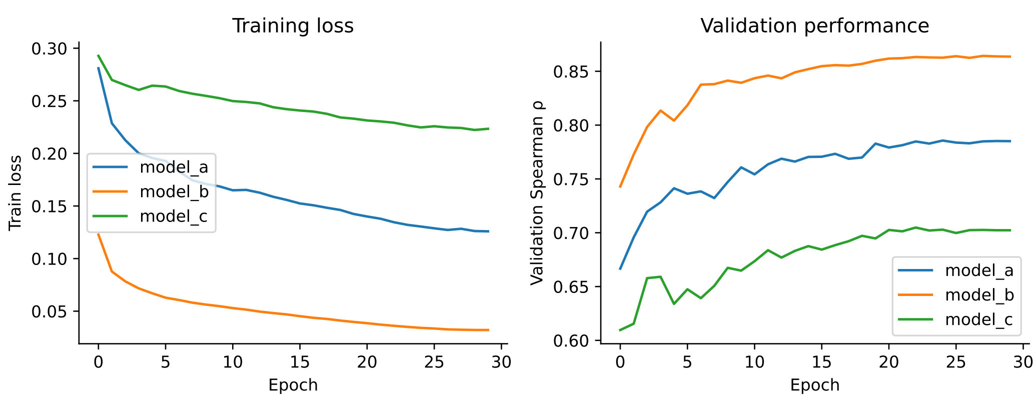

Figure 1: Training loss and validation Spearman ρ for the three initial scalar-head configurations: Model A (rank loss on), Model B (rank loss off), Model C (no context conditioning). Model B's training loss falls $\sim 4\times$ while validation performance plateaus after epoch $\sim$20 – the overfitting signal that motivated the simplex-head redesign (§3.1).

6 Head-to-Head with OptiPrime

6.1 Protocol

Every element of this comparison is matched. Both models train on OptiPrime's exact 297,962 training rows. Both use the same five cross-validation splits: model k holds out split k for validation and early stopping, and the five are ensembled, so each ensemble collectively covers all training data – exactly what OptiPrime's five released checkpoints do. Both are evaluated on the complete 20,509-row held-out set, which *neither* model has trained on, and which spans both studies newly generated or reprocessed for this comparison: 9,175 Liu rows and 11,334 Kim rows.

Two earlier confounds are eliminated. First, the held-out rows are in OptiPrime's native input format – verified for the Liu rows (protospacer at `PS20_OFFSET` = 4 on all 9,175) and inherited automatically for Kim, whose files are handed to OptiPrime's own loader unmodified – so no padding is applied and OptiPrime's PAM and seed-window features read real sequence (contrast §7). Second, because these are OptiPrime's own held-out rows, neither model is scored on data it trained on.

6.2 Result on the complete held-out set

Table 4: Head-to-head on the complete held-out set (n = 20,509). Both models are five-model cross-validation ensembles trained on the identical 297,962 rows. Best value per column in bold.

Model	Pearson r	Spearman ρ	MAE	RMSE
OptiPrime (official 5-model)	0.8270	0.8690	0.0590	0.1026
PE-RankFormer (5-model CV)	**0.8462**	**0.8865**	**0.0520**	**0.0963**

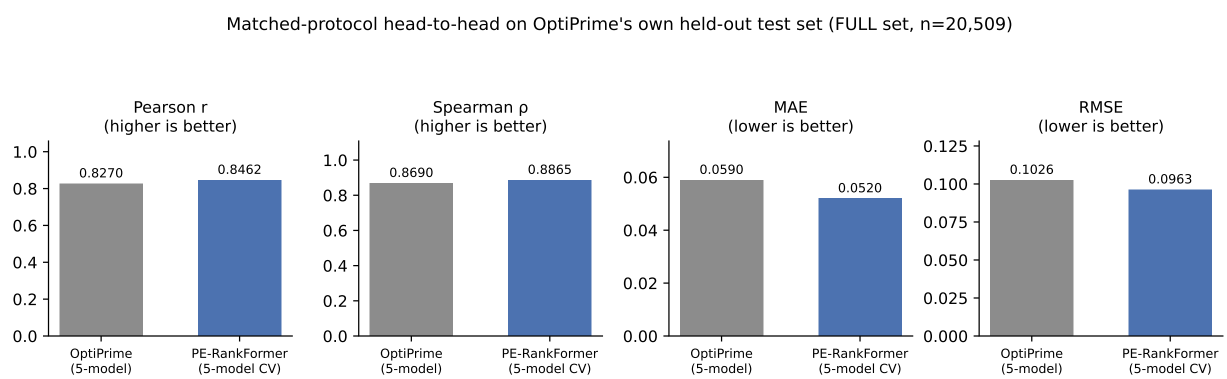

Figure 2: All four metrics, complete held-out set. PE-RankFormer leads on every one.

A paired, protospacer-clustered bootstrap (2000 resamples, 750 clusters) gives an observed difference of +0.0175, bootstrap mean +0.0173, 95% CI $[+0.0068, +0.0279]$, PE-RankFormer ahead in 100% of resamples, $p = 0.001$.

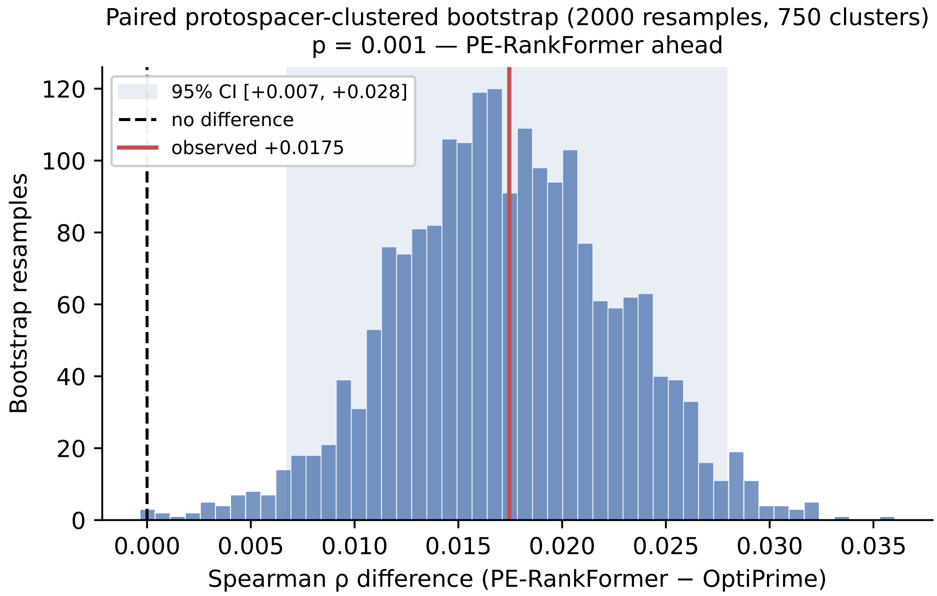

Figure 3: Bootstrap distribution of the paired Spearman difference, complete held-out set. The interval is clear of zero.

6.3 The result is not uniform: Liu vs. Kim

The pooled advantage is not evenly distributed. Splitting by source study:

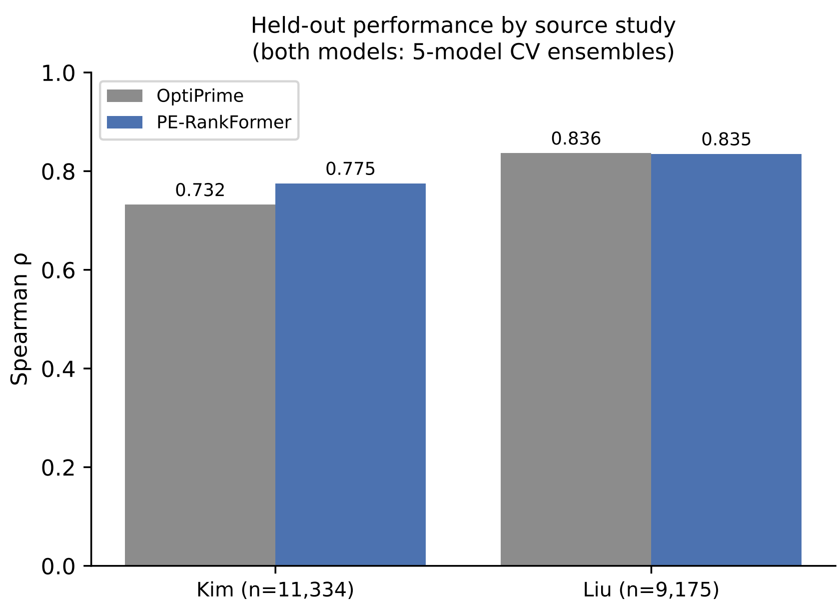

Figure 4: Spearman ρ by source study. Liu: a dead heat. Kim: PE-RankFormer ahead by 0.04.

Table 5: Head-to-head by source study, both 5-model CV ensembles.

Study	n	OptiPrime ρ	PE-RankFormer ρ
Liu (Hsu 2026)	9,175	0.8365	0.8349
Kim (DeepPrime)	11,334	0.7320	0.7751

On Liu, a paired protospacer-clustered bootstrap gives -0.0016, 95% CI $[-0.0249, +0.0218]$, $p = 0.94$ (150 clusters) – **statistically indistinguishable**, and this is the partition the paper's own reported figure of $\rho = 0.775$ refers to (§6.4). On Kim, breaking down further by the 18 individual experimental conditions, PE-RankFormer is ahead in 16 of 18, by margins of 0.02–0.09, with no single condition driving the effect: mean per-condition ρ is 0.760 for PE-RankFormer against 0.728 for OptiPrime.

We verified this is not a metadata artifact on our side: the Kim held-out files are handed to OptiPrime's own `scripts/pe/pe_datasets.py` unmodified, and its NRCH/Cas9-PAM inference from filename fires correctly for exactly the 4 of 18 Kim conditions that need it (checked against the corpus's own `NRCH` flag). The Kim advantage spans seven cell lines (A549, DLD1, HCT116, HEK293T, HeLa, MDA-MB-231, NIH3T3) and both PE2/PE4 systems, which is consistent with – though does not prove – an explanation grounded in architecture: Kim's greater experimental diversity is exactly the kind of context OptiPrime's FiLM-free design was not built to condition on, whereas PE-RankFormer's context encoder was designed for it (§5 shows removing it costs 0.08 Spearman on a different corpus). This is offered as a hypothesis, not a demonstrated cause.

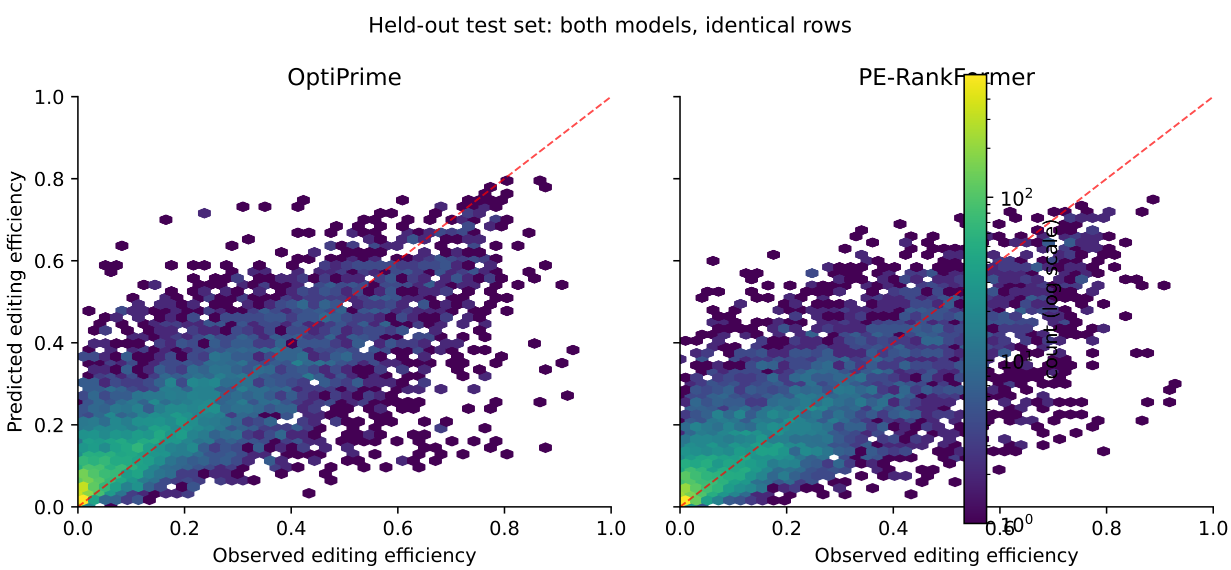

Figure 5: Predicted vs. observed efficiency for both models on the identical Liu held-out rows (log-scaled density).

6.4 Reproduction check against the published figure

The paper reports, for its own held-out test set, *"the cross-validation results translated well to the test set across all four experimental conditions tested in this work (mean $r = 0.723$, $\rho = 0.775$)"* – the four conditions being the Liu cell-type×PE-system combinations. Averaging our OptiPrime reproduction the same way (per-condition mean, not pooled) gives $r = 0.752$, $\rho = 0.805$: about 0.03 above the published value on both metrics.

We could not identify the source of this offset. The repository releases no evaluation script, only training code, and neither the main text nor Supplementary Text 1 specifies whether the published figure uses the released five-model ensemble, a single model trained on all data, or a sampled subset of the held-out rows (the text describes the sets as "sampled," without further detail). We checked every aggregation convention we could construct – per-model mean, ensemble mean, pooled, read-depth weighted, grouped by library instead of cell type – and none reproduces $(0.723, 0.775)$ jointly; the closest match, on both value and on the $\rho - r$ gap specifically, is the per-condition ensemble mean above.

The offset runs in OptiPrime's favor: our reproduction scores it higher than its own paper, not lower, so the baseline is not being handicapped. But it sets a floor on precision. A $\sim$0.03 reproduction uncertainty is roughly twenty times the Liu-only model difference (-0.0016) and about twice the full-set difference ($+0.0175$). The full-set result should be read as "PE-RankFormer is ahead by an amount larger than our reproduction uncertainty," not as a precise point estimate.

6.5 The cost of an unmatched protocol

An intermediate run makes the value of exact protocol matching concrete. Training three seeds on splits 2–5 – holding out split 1 for validation, hence using 80% of the available training rows against OptiPrime's effective 100% – gives Liu-only Spearman 0.8245, a deficit of -0.0120 (CI $[-0.035, +0.013]$, $p = 0.33$). Moving to the matched five-fold protocol recovers essentially all of that gap. The difference between the two runs is not modelling but bookkeeping, and on the Liu partition it is comparable in size to the effect one would be tempted to interpret.

7 A Retracted Result, and Why It Was Wrong

An earlier version of this report claimed a decisive win over OptiPrime: Spearman 0.807 against 0.724 on a 15,022-row test fold drawn from our public-data reconstruction, a margin of $+0.083$ with a bootstrap CI of $[+0.047, +0.118]$, $p < 0.0005$, and PE-RankFormer ahead in 2000 of 2000 resamples. A later run on an expanded corpus reproduced the same qualitative finding at $+0.070$. Both are withdrawn.

The cause was input preprocessing, not modelling. OptiPrime's `format_pe_df` requires the protospacer at a fixed offset of 4 within `full_unedited`, but the public Hsu supplementary workbook provides no upstream genomic context for the Lib-MMR designs, whose protospacers begin at offset 0. To run the baseline at all we left-padded those sequences with filler `'A'` bases. OptiPrime's PAM-proximal and seed-window features then read fabricated sequence for a majority of rows.

The error was not for want of checking. We ran a control splitting the test set by whether padding had been applied and found the two subsets scored almost identically, and concluded padding was not distorting the baseline. That control was too weak: it compared padded rows against a small, non-random unpadded remainder rather than against the same rows unpadded, which was impossible without the data we did not have. The direct comparison is now available and is unambiguous – on correctly formatted rows OptiPrime scores 0.8365 rather than 0.724.

Table 6: OptiPrime's measured performance depends far more on input formatting than the model difference under study.

Evaluation	Input condition	Spearman ρ
Our reconstructed test fold ($n = 15{,}022$)	filler-padded	0.7242
OptiPrime's held-out test set ($n = 9{,}175$)	native format	**0.8365**
Effect of padding artifact		≈ 0.11
Model difference on this partition (Liu)		0.0016
Model difference, complete held-out set		0.0175

The artifact was roughly seventy times the size of the Liu-partition model difference, and over six times the complete-held-out-set difference reported in §6. Two lessons generalise beyond this project. First, a baseline that must be reformatted to run is a baseline that can be silently handicapped, and aggregate metrics will not reveal it – 0.724 is a perfectly plausible score. Second, when a reimplementation beats a published model by a wide margin on the published model's own benchmark, the first hypothesis should be that the baseline is mis-run.

We report the retraction in full rather than quietly substituting the corrected number, since the withdrawn result is more instructive than the one that replaced it.

8 Discussion and Caveats

What the results do and do not license. On Liu, the bootstrap interval $[-0.025, +0.022]$ excludes any difference larger than about 0.025 Spearman in either direction, supporting parity specifically on that benchmark. On the complete held-out set the interval $[+0.007, +0.028]$ excludes zero, supporting a real but modest advantage. Neither generalises past what was tested: 20,509 rows from two studies do not establish behaviour on other edit types, cell lines, or efficiency regimes, and the Kim advantage in particular rests on a hypothesis about context-conditioning (§6) that was not directly tested by an ablation on the official corpus.

Reproduction uncertainty bounds precision, not direction. §6.4 found our OptiPrime reproduction running $\sim$0.03 Spearman above the paper's own published Liu-only figure, under every aggregation convention checked. This is a real limit: it means the complete-held-out-set difference (+0.0175) should be read as "an edge larger than our measurement uncertainty," not as a number precise to its fourth decimal. It does not undermine the qualitative comparisons (parity on Liu; an edge on Kim), since the offset applies equally to both models and moves with the input format, not with which model is being scored – but it is the reason we do not claim a precise magnitude for the pooled advantage.

Ensemble asymmetries are now resolved, and mattered. Earlier runs compared a 3-model ensemble trained on 80% of the data against a 5-model ensemble trained on all of it. That asymmetry alone accounted for a -0.0120 deficit — larger than the -0.0016 difference that remains under matched protocol. Any comparison of this kind should match ensemble size and training-data coverage before interpreting a gap.

Indel supervision is uneven. 42.5% of the official corpus has no measured indel, concentrated entirely in the Kim rows and Schwank diverse libraries. The simplex head marginalises rather than imputes for those rows, but it means the head's distinguishing signal is available for only part of the training data – including none of the Kim data. The gains attributed to the simplex head in §5 were measured on the public reconstruction, where indel coverage was complete; that ablation has not been rerun on the official mix.

Superseded ablations. The configuration sweep in §5 and the full-test-fold metrics were produced on the public-data reconstruction, using fold assignments that are not OptiPrime's. Their relative conclusions (simplex head helps; context conditioning helps; the ranking loss as configured hurts) are the basis for the architecture evaluated above, but the absolute numbers are not comparable to Table 4 and should not be quoted alongside it.

Calibration at high efficiency. Both models under-predict in the high-efficiency tail (Figure 5). This matters for downstream uses needing accurate absolute predictions above $\sim$80% efficiency, as opposed to relative ranking, where top-k regret is low.

9 Conclusion

Given OptiPrime's exact training data, its cross-validation splits, and its complete held-out test set, a minimally-mechanistic relational Transformer is ahead of a strongly mechanistic model by a margin outside chance: Spearman 0.8865 against 0.8690 on 20,509 rows neither model trained on, +0.0175 with a 95% CI of $[+0.0068, +0.0279]$ and $p = 0.001$.

That headline understates an important structure in the result. On the 9,175-row Liu partition – the specific benchmark the paper itself reports – the two models are statistically indistinguishable (-0.0016, $p = 0.94$). The entire measured advantage is concentrated in the 11,334-row Kim partition, which is more experimentally diverse (seven cell lines, multiple PE systems) than Liu. We regard both findings as load-bearing: parity on the paper's own primary benchmark, and an edge on a broader one, is a more precise and more defensible claim than either number alone, and considerably more precise than the retracted result below.

For the original research question – whether domain mechanism can be substituted for by scale plus a sufficiently expressive general-purpose architecture – the answer is a qualified yes, with a plausible mechanism for where the substitution pays off. OptiPrime's mechanistic prior is not costing it anything on the conditions closest to its own training distribution, and a general architecture with explicit context conditioning appears to generalise somewhat better as experimental diversity increases. That last clause is a hypothesis motivated by the data, not a demonstrated causal claim, and is flagged as such in §6.

The most consequential finding was methodological, not architectural. Our earlier reported margin of +0.070 to +0.083 Spearman, statistically significant at $p < 0.0005$ and stable across two corpus reconstructions, was an artifact of feeding the baseline filler-padded sequences it was never designed to receive – roughly seventy times the size of the Liu-partition model difference and six times the complete-set difference that replaced it. The artifact survived a control check we believed adequate at the time, and was detectable only once the original data became available. We record it in full, because the result that had to be withdrawn is more instructive than the one that replaced it.